

Networking concepts : DNS, IP, subnets & ports.

1. DNS - How names become IPs
What happens when you type google.com in browser?

- • Browser checks local cache for IP
- If not found, it queries a DNS resolver
- Resolver contacts root → TLD → authoritative DNS server
- Returns IP address, then browser connects to that IP.

• DNS record types -

- 1) A → Maps domain to IPv4 address
- 2) AAAA → Maps domain to IPv6 address
- 3) CNAME → Alias for another domain.
- 4) MX → Mail server for domain
- 5) NS → Name servers for domain.

dig google.com → Queries DNS records for domain.

2> IP Addressing -

- What is an IPv4 address?

→ A 32-bit numeric address written as 4 octets
e.g. 192.168.1.10

Used to identify device in the network.

- Public vs Private IP

Public IP → accessible over internet.

e.g. - 8.8.8.8

Private IP → used inside local networks
e.g. - 192.168.1.5

Private IP ranges -
10.0.0.0 - 10.255.255.255
172.16.0.0 - 172.31.255.255
192.168.0.0 - 192.168.255.255

• ip addr show → displays all network interface IPs.

3. CIDR & Subnetting

• What does /24 mean?
→ /24 → first 24 bits are network bits
remaining 8 bits are for hosts.

• No. of Hosts -

/24 → 256 total, 254 usable

/16 → 65536 total, 65534 usable

/28 → 16 total, 14 usable

• Why do we subnet?

→ To divide networks into smaller parts.

To improve security & performance
efficiency IP usage.

CIDR

Subnet mask

/24

255.255.255.0

/16

255.255.0.0

/28

255.255.255.240

4. Ports - The Doors to service

• What is port?

→ A logical endpoint to machine
 Helps identify which service should receive traffic.

Common Ports

Services

22

SSH

80

HTTP

443

HTTPS

53

DNS

3306

MySQL

6379

Redis

27017

MongoDB

5. Putting it together

• curl http://myapp.com:8080 → what happens?

DNS resolves myapp.com → IP

TCP connection established on port 8080

HTTP request sent to server.

• App can't reach DB at 10.0.1.50:3306 →

then check -

Network connectivity (ping)

Port open (ss /nc)

firewall rules

DB service running